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Prodi : Sistem Informasi

Izin menjawab diskusi 6

Soal

1. **Diberikan matriks sebagai berikut:**

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 2 & 1 & 0 \end{pmatrix}$$

Tentukan invers matriks dengan metode adjoint

2. **Berikan contoh sistem persamaan linear dua variabel, kemudian selesaikan sistem tersebut menggunakan aturan Cramer**

Jawaban

1. Dik matriks: $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 2 & 1 & 0 \end{pmatrix}$

   Rumus invers matriks dengan metode matriks adjoint yang ada pada materi inisiai sesi 6 STMA4113 yaitu : $A^{-1} = \frac{1}{\det(A)} adj(A)$

   a. Menentukan Kofaktor Matriks (A)

$$C_{11} = + \begin{pmatrix} 1 & 3 \\ 1 & 0 \end{pmatrix} = (1 \cdot 0) - (3 \cdot 1) = -3$$

$$C_{12} = + \begin{pmatrix} 0 & 3 \\ 2 & 0 \end{pmatrix} = -((0 \cdot 0) - (3 \cdot 2)) = -(-6) = 6$$

$$C_{13} = + \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} = (0 \cdot 1) - (1 \cdot 2) = -2$$

$$C_{21} = + \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} = -\big((2 \cdot 0) - (1 \cdot 1)\big) = -(-1) = 1$$

$$C_{22} = + \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} = (1 \cdot 0) - (1 \cdot 2) = -2$$

$$C_{23} = + \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = -\big((1 \cdot 1) - (2 \cdot 2)\big) = -3(-3) = 3$$

$$C_{31} = + \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = (2 \cdot 3) - (1 \cdot 1) = 6 - 1 = 5$$

$$C_{32} = + \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} = -\big((1 \cdot 3) - (1 \cdot 0)\big) = -3$$

$$C_{33} = + \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = (1 \cdot 1) - (2 \cdot 0) = 1$$

Kofaktor $A = \begin{pmatrix} -3 & 6 & -2 \\ 1 & -2 & 3 \\ 5 & -3 & 1 \end{pmatrix}$

b. Menghitung Determinan Matriks A (det(A))

$$\det(A) = \begin{pmatrix} 1 & 2 & 1 & | & 1 & 2 \\ 0 & 1 & 3 & | & 0 & 1 \\ 2 & 1 & 0 & | & 2 & 1 \end{pmatrix}$$

$$\det(A) = \big((1 \cdot 1 \cdot 0) + (2 \cdot 3 \cdot 2) + (1 \cdot 0 \cdot 1)\big) - ((2 \cdot 1 \cdot 1) + (1 \cdot 3 \cdot 1) + (0 \cdot 0 \cdot 2))$$

$$\det(A) = (0 + 12 + 0) - (2 + 3 + 0)$$

$$\det(A) = 12 - 5 = 7$$

$$adj(A) = \begin{pmatrix} -3 & 1 & 5 \\ 6 & -2 & -3 \\ -2 & 3 & 1 \end{pmatrix}$$

$$A^{-1} = \frac{adj(A)}{(A)} = \frac{1}{7}\begin{pmatrix} -3 & 1 & 5 \\ 6 & -2 & -3 \\ -2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{3}{7} & \frac{1}{7} & \frac{5}{7} \\ \frac{6}{7} & -\frac{2}{7} & -\frac{3}{7} \\ -\frac{2}{7} & \frac{3}{7} & \frac{1}{7} \end{pmatrix}$$

2. Contoh Persamaan Linear Dua Variabel:

$$\begin{Bmatrix} 2_x + 3_y = 12 \\ 3_x - y = 7 \end{Bmatrix}$$

Mengubah bentuk persamaan matriks $AX = B$:

$$\begin{pmatrix} 2 & 3 \\ 3 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 12 \\ 7 \end{pmatrix}$$

a. Menghitung Determinan Variabel (D)

$$D = \begin{pmatrix} 2 & 3 \\ 3 & -1 \end{pmatrix} = (2 \cdot -1) - (3 \cdot 3) = -2 - 9 = -11$$

b. Menghitung Determinan Variabel $x(D_x)$

$$D_x = \begin{pmatrix} 12 & 3 \\ 7 & -1 \end{pmatrix} = (12 \cdot -1) - (3 \cdot 7) = -12 - 21 = -33$$

c. Menghitung Determinan Variabel $y(D_y)$

$$D_y = \begin{pmatrix} 2 & 12 \\ 3 & 7 \end{pmatrix} = (2 \cdot 7) - (12 \cdot 3) = 14 - 36 = -22$$

d. Menentukan nilai $x\ dan\ y$

$$x = \frac{D_x}{D} = \frac{-33}{-11} = 3$$

$$y = \frac{D_y}{D} = \frac{-22}{-11} = 2$$

Himpunan penyelesaiannya adalah {(3,2)}